

EQUATIONS REDUCIBLE TO A PAIR OF LINEAR EQUATIONS

We have several situations when there are two equations that are not linear, but can be reduced to a pair of linear equations. By making suitable substitutions.

EXAMPLE 7. If $\frac{2}{x} + \frac{3}{y} = 13$ and $\frac{5}{x} - \frac{4}{y} = -2$, then

what is the value of $\frac{x}{y}$?

- a. $\frac{2}{3}$ b. $\frac{3}{2}$ c. $\frac{1}{3}$ d. $\frac{1}{2}$

Sol. b. Let $\frac{1}{x} = p$ and $\frac{1}{y} = q$

So, we have, $2p + 3q = 13$... (i)

and $5p - 4q = -2$... (ii)

On multiplying Eq. (i) by 4 and Eq. (ii) by 3 and then adding, we get

$$8p + 15p = 52 - 6 \Rightarrow 23p = 46 \Rightarrow p = 2$$

On putting $p = 2$ in Eq. (i), we get

$$2 \times 2 + 3q = 13 \Rightarrow 3q = 13 - 4 = 9 \Rightarrow q = \frac{9}{3} = 3$$

Now, $x = \frac{1}{p} \Rightarrow x = \frac{1}{2}$ and $y = \frac{1}{q} \Rightarrow y = \frac{1}{3}$.

Hence, the value of $\frac{x}{y}$ is $\frac{3}{2}$.

EXAMPLE 8. Find the solution of the system of equations $4x + 3y = 18xy$ and $2x - 5y = -4xy$.

- a. $x = 2, y = 3$ b. $x = 3, y = 2$ c. $x = \frac{1}{3}, y = \frac{1}{2}$ d. $x = \frac{1}{2}, y = \frac{1}{3}$

Sol. d. Given equations are, $4x + 3y = 18xy$... (i)

and $2x - 5y = -4xy$... (ii)

On dividing both the equations by xy , we get

$$\frac{4x}{xy} + \frac{3y}{xy} = \frac{18xy}{xy} \text{ and } \frac{2x}{xy} - \frac{5y}{xy} = \frac{-4xy}{xy}$$

or $\frac{4}{y} + \frac{3}{x} = 18$ and $\frac{2}{y} - \frac{5}{x} = -4$

On putting $\frac{1}{y} = A$ and $\frac{1}{x} = B$, we get

$$4A + 3B = 18 \quad \dots \text{(iii)}$$

$$2A - 5B = -4 \quad \dots \text{(iv)}$$

On multiplying Eq. (iv) by 2 and subtracting from Eq. (iii), we get

$$\begin{array}{r} 4A + 3B = 18 \\ 4A - 10B = -8 \\ \hline -7B = 26 \end{array}$$

$$\Rightarrow B = 2$$

On putting, the value of B in Eq. (iii), we get

$$\Rightarrow 4A + 6 = 18 \Rightarrow 4A = 12 \Rightarrow A = 3$$

$$\text{Now } \frac{1}{x} = B = 2 \Rightarrow x = \frac{1}{2} \text{ and } \frac{1}{y} = A = 3 \Rightarrow y = \frac{1}{3}$$

So, $x = \frac{1}{2}$ and $y = \frac{1}{3}$ is the required solution.

EXAMPLE 9. If $\frac{3}{x+y} + \frac{2}{x-y} = 2$ and $\frac{9}{x+y} - \frac{4}{x-y} = 1$,

then what is value of $\frac{x}{y}$?

- a. $\frac{3}{2}$ b. 5 c. $\frac{2}{3}$ d. $\frac{1}{5}$

Sol. b. Given equations are,

$$\frac{3}{x+y} + \frac{2}{x-y} = 2 \quad \dots \text{(i)}$$

$$\text{and } \frac{9}{x+y} - \frac{4}{x-y} = 1 \quad \dots \text{(ii)}$$

$$\text{Let } \frac{1}{x+y} = a \text{ and } \frac{1}{x-y} = b$$

$$\therefore 3a + 2b = 2 \quad \dots \text{(iii)}$$

$$9a - 4b = 1 \quad \dots \text{(iv)}$$

On multiplying Eq. (iii) by 2 and adding with Eq. (iv), we get

$$\Rightarrow 6a + 9a = 5 \Rightarrow 15a = 5$$

$$\therefore a = \frac{1}{3}$$

$$\therefore \frac{1}{x+y} = \frac{1}{3} \Rightarrow x+y = 3 \quad \dots \text{(v)}$$

On putting the value of a in Eq. (iii), we get

$$3 \times \frac{1}{3} + 2b = 2 \Rightarrow b = \frac{1}{2}$$

$$\Rightarrow \frac{1}{x-y} = \frac{1}{2} \Rightarrow x-y = 2 \quad \dots \text{(vi)}$$

On adding Eqs. (v) and (vi), we get, $2x = 5 \Rightarrow x = \frac{5}{2}$

$$\text{From Eq. (v), } x+y = 3 \Rightarrow y = 3 - \frac{5}{2} = \frac{1}{2}$$

$$\therefore \frac{x}{y} = \frac{\frac{5}{2}}{\frac{1}{2}} = 5$$

LINEAR EQUATION IN THREE VARIABLES

An equation in the form of $ax + by + cz = r$, where a, b, c and r are real numbers and a, b and c are not all zeros is called a linear equation in three variables

e.g. $3x + 4y - 7z = 2$, $-2x + y - z = 6$ and $x + y + z = 2$ are all linear equation in three variables.

Use substitution and elimination method to solve the system of three equations in three variables.

EXAMPLE 10. Solve the system $x + y + z = 5$,
 $2x - y + z = 9$ and $x - 2y + 3z = 16$.

a. (2, -1, 4) b. (3, 5, 6) c. (1, -2, 5) d. (6, 1, -3)

Sol. a. Given equations are, $x + y + z = 5$... (i)
 $2x - y + z = 9$... (ii)
 $x - 2y + 3z = 16$... (iii)

First we eliminate y by adding Eqs. (i) and (ii), we get

$$\begin{array}{r} x + y + z = 5 \\ 2x - y + z = 9 \\ \hline 3x + 2z = 14 \end{array} \quad \dots (iv)$$

Next we eliminate y by multiplying Eq. (i) by 2 and then adding it to Eq. (iii), we get

$$\begin{array}{r} 2x + 2y + 2z = 10 \\ x - 2y + 3z = 16 \\ \hline 3x + 5z = 26 \end{array} \quad \dots (v)$$

Now subtract Eq. (iv) from Eq. (v), we get

$$3z = 12 \Rightarrow z = 4$$

Substitute $z = 4$ in Eq. (iv), $3x + 2 \times 4 = 14 \Rightarrow 3x = 6 \Rightarrow x = 2$.

Finally put $x = 2$ and $z = 4$ in Eq. (i), we get

$$2 + y + 4 = 5 \Rightarrow y = -1$$

$\therefore (2, -1, 4)$ is the required solution.

APPLICATION OF LINEAR EQUATIONS

The application of linear equations are discussed below.

Problem Based on Ages

If the problem involves finding out the ages of two persons, then take the present age of one person as x and that of the other as y .

Then, ' a ' years ago, age of 1st person was $(x - a)$ years and that of 2nd person was $(y - a)$ years.

After b years, age of 1st person will be $(x + b)$ years and that of 2nd person will be $(y + b)$ years.

Formulate the equations and then solve them.

EXAMPLE 11. Age of X is six times that of Y . After 4 yr, X is 4 times elder to Y . What is the present age of Y ?

a. 4 yr b. 5 yr c. 6 yr d. 7 yr

Sol. c. Let the present age of X and Y be x and y yr, respectively.

Condition I Age of X = Age of $Y \times 6$, $x = 6y$... (i)

After 4 yr, Age of $X = (x + 4)$ yr, Age of $Y = (y + 4)$ yr

Condition II $(x + 4) = (y + 4) \times 4$, $x + 4 = 4y + 16$... (ii)

Put the value of $x = 6y$ in Eq. (ii), we get

$$\Rightarrow 6y + 4 = 4y + 16 \Rightarrow 2y = 12$$

$$\therefore y = 6$$

So, the present age of Y is 6 yr.

Problem Based on Numbers

Let the digit in unit's place be x and that in ten's place be y . Then, the two-digits number is given by $10y + x$. On interchanging the positions of the digits, the digit in unit place becomes y and that in ten's place becomes x , and the number becomes $10x + y$. Formulate the equations and then solve them.

EXAMPLE 12. A number consists of two digits, whose sum is 8. If 18 is added to the number, the digits are reversed. The number is equal to

a. 26 b. 35 c. 53 d. 62

Sol. b. Let the unit's place digit and ten's place digit be x and y , respectively.

$\therefore$ Original number = $10y + x$

Number obtained by reversing the order of digit = $10x + y$

Condition I Sum of digits = 8

$$\Rightarrow x + y = 8 \quad \dots (i)$$

Condition II $10x + y = 10y + x + 18$

$$\Rightarrow 10x - 10y + y - x = 18 \Rightarrow 9x - 9y = 18$$

$$\Rightarrow x - y = 2 \quad \dots (ii)$$

On adding Eqs. (i) and (ii), we get

$$x + y + x - y = 8 + 2 \Rightarrow 2x = 10$$

$$\therefore x = 5$$

On putting $x = 5$ in Eq. (i), we get

$$5 + y = 8 \Rightarrow y = 3$$

Hence, Required number = $10y + x = 10 \times 3 + 5 = 35$

Problem Based on Fractions

Let the numerator of the fraction be x and denominator be y , then the fraction is x/y .

Formulate the linear equations on the basis of conditions given and solve them.

EXAMPLE 13. If we add 1 to the numerator and subtract 1 from the denominator, a fraction becomes

1. It also becomes $\frac{1}{2}$, if we only add 1 to the denominator. The fraction is

a. 4/5 b. 3/5 c. 2/5 d. 1/5

Sol. b. Let the numerator and the denominator of the fraction be x and y , respectively.

$$\therefore \text{Fraction} = \frac{x}{y}$$

Condition I When 1 is added to the numerator and 1 is subtract from the denominator, then new fraction = $\frac{x+1}{y-1}$

According to the question,

$$\frac{x+1}{y-1} = 1 \Rightarrow x+1 = y-1 \Rightarrow x-y = -2 \quad \dots (i)$$

Condition II When 1 is added to denominator, then new fraction = $\frac{x}{y+1}$

According to the question, $\frac{x}{y+1} = \frac{1}{2} \Rightarrow 2x = y + 1$
 $\Rightarrow 2x - y = 1 \quad \dots(ii)$

On subtracting Eq. (ii) from Eq. (i), we get

$$-x = -3 \Rightarrow x = 3$$

Put the value of x in Eq. (i), we get

$$3 - y = -2 \Rightarrow y = 5$$

Hence, the fractions is $\frac{x}{y} = \frac{3}{5}$

Problem Based on Distance, Speed and Time

Speed is the distance covered by an object per unit time.

i.e. $\text{Speed} = \frac{\text{Distance}}{\text{Time}}$

If the speed of boat in still water be x km/h and speed of stream be y km/h. Then, the speed of boat downstream = $(x + y)$ km/h and speed of boat upstream = $(x - y)$ km/h

EXAMPLE 14. A motorboat takes 2h to travel a distance of 9 km down the current and it takes 6h to travel the same distance against the current. What is the speed of the boat in still water ?

a. 3 km/h b. 2 km/h c. 1.5 km/h d. 1 km/h

Sol. a. Let speed of the motorboat in still water be x km/h and speed of the stream be y km/h.

Then, speed of the motorboat downstream = $(x + y)$ km/h and speed of the boat upstream = $(x - y)$ km/h

Condition I When motorboat goes 9 km downstream, then

$$2 = \frac{9}{x+y} \quad \left[\because \text{Time} = \frac{\text{Distance}}{\text{Speed}} \right]$$

$$\Rightarrow 2x + 2y = 9 \quad \dots(i)$$

Condition II When motorboat goes 9 km upstream, then

$$6 = \frac{9}{x-y} \Rightarrow 6x - 6y = 9 \quad \dots(ii)$$

On multiplying Eq. (i) by 3, we get

$$6x + 6y = 27 \quad \dots(iii)$$

Now, adding Eqs. (ii) and (iii), we get

$$12x = 36 \Rightarrow x = 3 \text{ km/h}$$

Problem Based on Fixed and running Charges

Let the fixed charge of any commodity be ₹ x and rate of running charges be ₹ y , then total cost = x + total running charges.

Formulate the equations and then solve them.

EXAMPLE 15. A railway ticket for a child costs half the full fare but the reservation charge is the same on half tickets as much as on full ticket. One reserved first class ticket for a journey between two stations is ₹ 362, one full and one half reserved first class tickets cost ₹ 554. What is the reservation charge?

a. ₹ 18 b. ₹ 22 c. ₹ 38 d. ₹ 46

Sol. b. Let full fare and reservation charge be ₹ x and ₹ y , respectively. Then, full ticket = $x + y$

$$\text{half ticket} = \frac{x}{2} + y$$

$$\text{According to question, } x + y = 362 \quad \dots(i)$$

$$\text{and } (x + y) + \left(\frac{x}{2} + y \right) = 554 = \frac{3}{2}x + 2y = 554 \quad \dots(ii)$$

On multiplying Eq. (i) by 2, we get

$$2x + 2y = 724 \quad \dots(iii)$$

On subtracting Eq. (ii) from Eq. (iii), we get

$$\frac{x}{2} = 170 \Rightarrow x = 340$$

On putting value of x in Eq. (i), we get

$$340 + y = 362 \Rightarrow y = 22$$

Hence, the reservation charge is ₹ 22.

Problems Based on Mensuration

Make use of given formulae to formulate the equation

- Area of rectangle = length $\times$ breadth
Perimeter of rectangle = 2(length+Breadth)
- The sum of angles of a triangle is 180°
- The sum of opposite angles of cyclic quadrilateral is 180°
- In parallelogram, opposite angles are equal.

EXAMPLE 16. The length of a rectangle is 8 cm more than its breadth. If the perimeter of the rectangle of 68 cm, then its length and breadth are

a. 21 cm, 13 cm b. 14 cm, 23 cm
c. 19 cm, 20 cm d. 9 cm, 15 cm

Sol. a. Let the length and breadth of a rectangle be x and y , respectively.

$$\text{Condition I } x = y + 8 \Rightarrow x - y = 8 \quad \dots(i)$$

Condition II Perimeter of rectangle = 68 cm

$$\Rightarrow 2[x + y] = 68, \quad x + y = 34 \quad \dots(ii)$$

On adding Eq. (i) and (ii), we get

$$2x = 42 \Rightarrow x = 21$$

On putting $x = 21$ in Eq. (ii), we get

$$21 + y = 34 \Rightarrow y = 34 - 21 = 13$$

Hence, length of rectangle = 21 cm

and breadth of rectangle = 13 cm

EXAMPLE 17. The area of a rectangle is decreased by 28 m^2 , if the length is increased by 2 m and the breadth is decreased by 2 m. The area of a rectangle is increased by 33 m^2 , if the length is decreased by 1 m and the breadth is increased by 2 m, then the area of rectangle is

- a. 253 m^2 b. 235 m^2 c. 532 m^2 d. 352 m^2

Sol. a. Let the length and breadth of a rectangle be x and y , respectively.

$\therefore$ Original area of rectangle $= xy \text{ m}^2$

Condition I If length $= (x + 2) \text{ m}$ and breadth $= (y - 2) \text{ m}$

$\therefore$ Area of rectangle $= (x + 2)(y - 2) \text{ m}^2$

According to the question,

$$(x + 2)(y - 2) = xy - 28$$

$$\Rightarrow xy - 2x + 2y - 4 = xy - 28$$

$$\Rightarrow 2x - 2y = 24 \Rightarrow x - y = 12 \quad \dots(i)$$

Condition II If, length $= (x - 1) \text{ m}$ and breadth $= (y + 2) \text{ m}$

$\therefore$ Now the area of rectangle $= (x - 1)(y + 2) \text{ m}^2$

According to the question,

$$(x - 1)(y + 2) = xy + 33$$

$$\Rightarrow xy + 2x - y - 2 = xy + 33$$

$$\Rightarrow 2x - y = 35 \quad \dots(ii)$$

On subtracting Eq. (i) from (ii), we get, $x = 23$

On putting $x = 23$ in Eq. (i), we get,

$$23 - y = 12$$

$$\therefore y = 23 - 12 = 11$$

So, original area of rectangle $= xy = 23 \times 11 = 253 \text{ m}^2$

PRACTICE EXERCISE

1. Find x , if $25x - 19 - [3 - \{4x - 5\}] = 3x - (6x - 5)$.

- (a) $x = 1$ (b) $x = -1$ (c) $x = \frac{1}{2}$ (d) $x = 2$

2. If a number is subtracted from three-fourth of itself, the value so obtained is -130 . Then, what is the number?

- (a) 540 (b) 560 (c) 420 (d) 520

3. Sum of two numbers is 21 and their difference is 11, then the greatest number is

- (a) 5 (b) 16 (c) 9 (d) 10

4. Which of the following equations have $x = 2$ and $y = 1$ as a solution?

I. $2x + 5y = 9$

II. $5x + 3y = 14$

III. $2x + 3y = 7$

IV. $2x - 3y = 1$

Select the correct answer using the codes given below

- (a) I and IV (b) II and III (c) Only I (d) I, III and IV

5. The solution of the pair of equation $\frac{x}{2} + y = 0.8$

and $\frac{7}{x + \frac{y}{2}} = 10$ is

(a) $x = \frac{2}{5}, y = \frac{3}{5}$

(b) $x = \frac{2}{3}, y = 5$

(c) $x = \frac{2}{5}, y = \frac{5}{3}$

(d) $x = \frac{3}{5}, y = \frac{2}{5}$

6. A system of two simultaneous linear equations in two variables has a unique solution if their graphs

- (a) are coincident (b) are parallel
(c) intersect in one point (d) None of these

7. The system of equations $6x + 5y = 11$ and $9x + \frac{15}{2}y = 21$ has

- (a) a unique solution (b) many solution
(c) no solution (d) None of these

8. The sum of two numbers is 2490 and if 6.5% of one number is equal to 8.5% of the other, then numbers are

- (a) 1414, 1076 (b) 1411, 1079
(c) 1412, 1078 (d) None of these

9. Given two linear equations $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$, if $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$, then the graph is

- (a) parallel (b) intersection at one point
(c) coincident (d) None of these

10. For what value of k , the following equations will be inconsistent? $4x + 6y = 11$ and $2x + ky = 7$

- (a) $k = -3$ (b) $k = \frac{12}{5}$ (c) $k = 12$ (d) $k = 3$

11. For what value of k , the system of equations has infinitely many solutions $2x - ky = 4$ and $3x + 2y = 6$?

- (a) $\frac{4}{3}$ (b) $-\frac{4}{3}$ (c) $\frac{2}{3}$ (d) $\frac{3}{2}$

12. For what value of α , the system of equation $\alpha x + 3y = \alpha - 3$, $12x + \alpha y = \alpha$ will have a unique solution?

- (a) $\alpha = \pm 6$ (b) $\alpha = 6$ (c) $\alpha \neq \pm 6$ (d) $\alpha = -6$